

1 Summary

Model (Run-name): model1 (test)

Iteration: 4

TAC: 145071.80721272656

Took: 0.40984511375427246s

Total Time: 1.198096513748169s

Epsilons:

- ε_{LMTD} : 1e-06
- $\varepsilon_{Balancing}$: 1e-06
- ε_{Area} : 1e-06
- ε_{Beta} : 1e-06

Gurobi Tolerances:

- IntFeasTol: 1e-05
- FeasibilityTol: 1e-06
- OptimalityTol: 1e-06

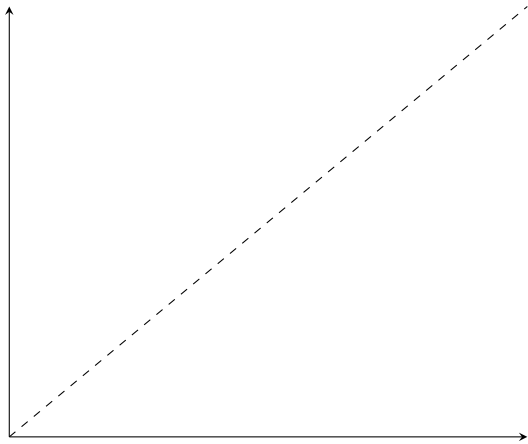
Gamsworld Solution: http://www.gamsworld.org/minlp/minlplib2/html/heatexch_gen1.html

2 Results

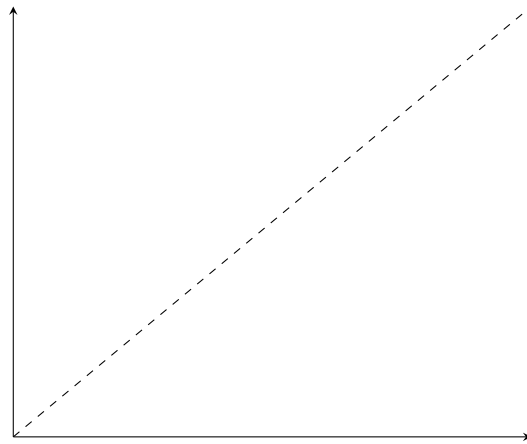
Variable	Value	Variable	Value	Variable	Value
$z_{1,1,1}$	1.0	$RecLMTD_{hu,1}$	0.021337898479317944	$b_{1,1,1}^{C,out}$	9048.226029948552
$z_{1,2,2}$	1.0	$q_{1,1,1}$	850.0	$b_{1,2,2}^{C,out}$	6500.0
$z_{2,1,2}$	1.0	$q_{1,2,2}$	1950.0	$b_{2,1,2}^{C,out}$	7389.980133572261
$z_{cu,2}$	1.0	$q_{2,1,2}$	2166.028659840146		
$z_{hu,1}$	1.0	$q_{cu,2}$	2233.971340159854		
$A_{1,1,1}^\beta$	69.14107665413653	$q_{hu,1}$	583.9713401598542		
$A_{1,2,2}^\beta$	84.1468297226002	$t_{1,1}^{(H)}$	650.0		
$A_{2,1,2}^\beta$	42.69900440382105	$t_{1,2}^{(H)}$	565.0		
$A_{cu,2}^\beta$	38.88483572219945	$t_{1,3}^{(H)}$	370.0		
$A_{hu,1}^\beta$	14.09178614751222	$t_{2,1}^{(H)}$	590.0		
$A_{1,1,1}$	69.14107665413653	$t_{2,2}^{(H)}$	590.0		
$A_{1,2,2}$	84.1468297226002	$t_{2,3}^{(H)}$	481.6985670079927		
$A_{2,1,2}$	42.69900440382105	$t_{1,1}^{(C)}$	611.0685773226764		
$A_{cu,2}$	38.88483572219945	$t_{1,2}^{(C)}$	554.4019106560097		
$A_{hu,1}$	14.09178614751222	$t_{1,3}^{(C)}$	410.0		
$dt_{1,1,1}$	38.93142267732361	$t_{2,1}^{(C)}$	500.0		
$dt_{1,1,2}$	10.598089343990296	$t_{2,2}^{(C)}$	500.0		
$dt_{1,2,2}$	65.0	$t_{2,3}^{(C)}$	350.0		
$dt_{1,2,3}$	20.0	$t_{1,1,1}^H$	370.0		
$dt_{2,1,2}$	35.5980893439903	$t_{1,2,2}^H$	370.0		
$dt_{2,1,3}$	71.69856700799271	$t_{2,1,2}^H$	481.6985670079927		
$dt_{cu,2}$	161.69856700799272	$t_{1,1,1}^C$	614.9952404843539		
$dt_{hu,1}$	68.93142267732361	$t_{1,2,2}^C$	500.0		
$f_{1,1,1}^H$	3.0357142857142856	$t_{2,1,2}^C$	580.0		
$f_{1,2,2}^H$	9.512857706269642	$b_{1,1,1}^{H,in}$	1973.2142857142858		
$f_{2,1,2}^H$	20.0	$b_{1,2,2}^{H,in}$	5469.757351319768		
$f_{1,1,1}^C$	14.712676512459526	$b_{2,1,2}^{H,in}$	11800.0		
$f_{1,2,2}^C$	13.0	$b_{1,1,1}^{H,out}$	1123.2142857142858		
$f_{2,1,2}^C$	12.741345057883208	$b_{1,2,2}^{H,out}$	3519.7573513197676		
$RecLMTD_{1,1,1}$	0.04553062065175502	$b_{2,1,2}^{H,out}$	9633.971340159855		
$RecLMTD_{1,2,2}$	0.02157611018528212	$b_{1,1,1}^{C,in}$	8198.226029948552		
$RecLMTD_{2,1,2}$	0.0160123216721665	$b_{1,2,2}^{C,in}$	4550.0		
$RecLMTD_{cu,2}$	0.00910851459890659	$b_{2,1,2}^{C,in}$	5223.951473732115		

3 *RecLMTD* Tangent Points

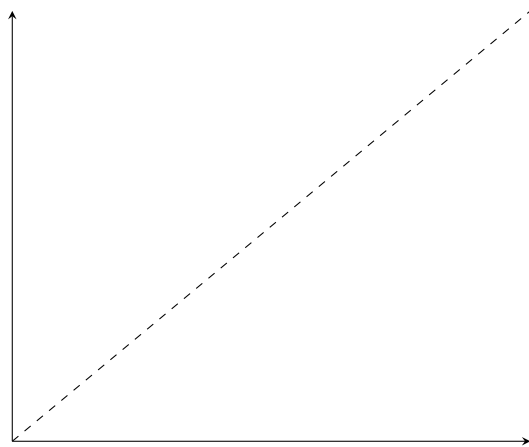
- $(1, 1, 1)$
 - Value: 0.045531
 - Error: 0.000391543035859
 - Relative error: 0.008526232311760
- $(1, 2, 2)$
 - Value: 0.021576
 - Error: 0.004616223066754
 - Relative error: 0.176243293116910
- $(2, 1, 2)$
 - Value: 0.016012
 - Error: 0.003382956187479
 - Relative error: 0.174421640770512
- $cu, 2$
 - Value: 0.009109
 - Error: 0.000021820691727
 - Relative error: 0.002389911326590
- $hu, 1$
 - Value: 0.021338
 - Error: 0.000030823680365
 - Relative error: 0.001442467178667



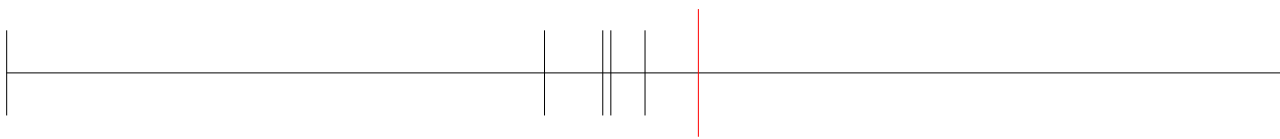
(a) *RecLMTD* - (1, 1, 1)



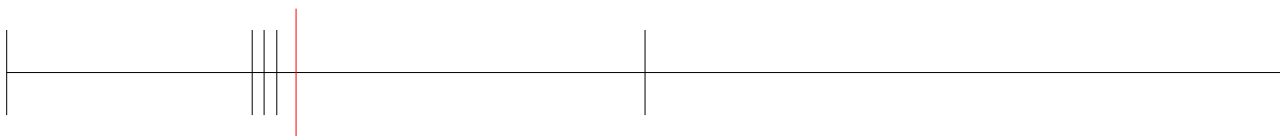
(b) *RecLMTD* - (1, 2, 2)



(c) *RecLMTD* - (2, 1, 2)



(a) *RecLMTD* - $cu, 2$



(a) *RecLMTD* - $hu, 1$

4 Balancing Breakpoints

4.1 $b^{H,\text{in}}$

- (1, 1, 1)
 - Value: 1973.2142857142858
 - Absolute error: 2.2737367544323206e-13
 - Relative error: 1.1523009796218097e-16
- (1, 2, 2)
 - Value: 5469.757351319768
 - Absolute error: 94.99274727741977
 - Relative error: 0.01767384327975516
- (2, 1, 2)
 - Value: 11800.0
 - Absolute error: 0.0
 - Relative error: 0.0

4.2 $b^{H,\text{out}}$

- (1, 1, 1)
 - Value: 1123.2142857142858
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 3519.7573513197676
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 9633.971340159855
 - Absolute error: 0.0
 - Relative error: 0.0

4.3 $b^{C,\text{in}}$

- (1, 1, 1)
 - Value: 8198.226029948552
 - Absolute error: 41.49006057719362
 - Relative error: 0.005086600906660372
- (1, 2, 2)
 - Value: 4550.0
 - Absolute error: 0.0

- Relative error: 0.0
- (2, 1, 2)
 - Value: 5223.951473732115
 - Absolute error: 0.0
 - Relative error: 0.0

4.4 $b^{C,out}$

- (1, 1, 1)
 - Value: 9048.226029948552
 - Absolute error: 0.0
 - Relative error: 0.0
- (1, 2, 2)
 - Value: 6500.0
 - Absolute error: 0.0
 - Relative error: 0.0
- (2, 1, 2)
 - Value: 7389.980133572261
 - Absolute error: 9.094947017729282e-13
 - Relative error: 1.2307133244393248e-16



(a) $t^{(H)} - (1, 1)$



(a) $t^{(H)} - (1, 2)$



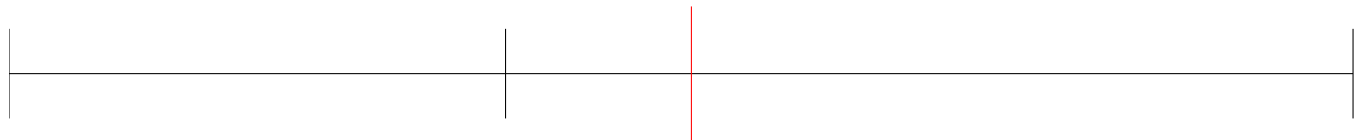
(a) $t^{(H)} - (2, 2)$



(a) $t^{(H)} - (1, 1, 1)$



(a) $t^{(H)} - (1, 2, 2)$



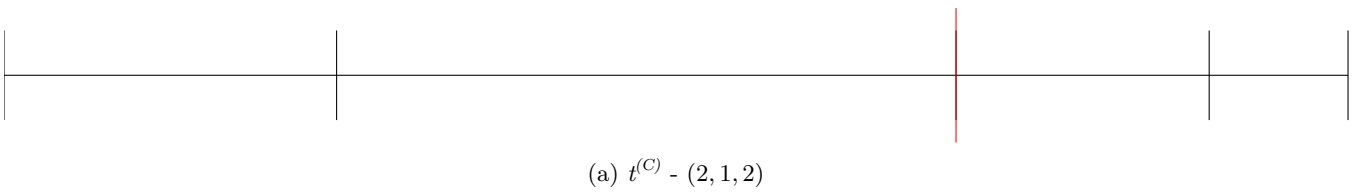
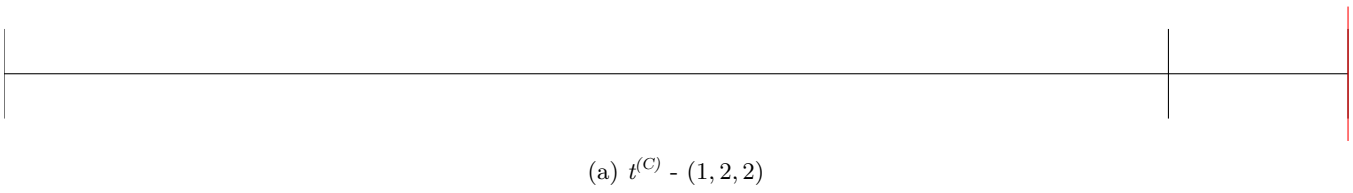
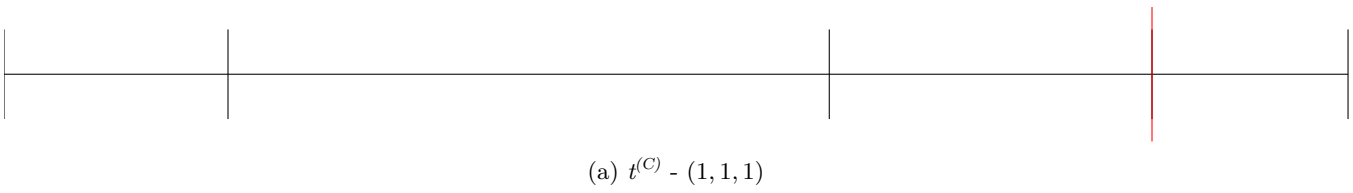
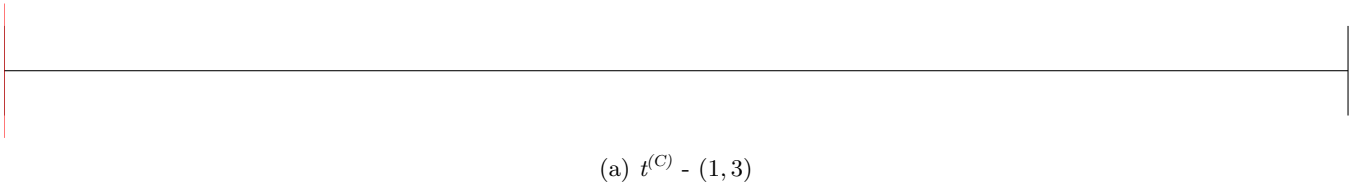
(a) $t^{(H)} - (2, 1, 2)$



(a) $t^{(C)} - (1, 2)$



(a) $t^{(C)} - (2, 3)$



5 Area Breakpoints

5.1 A

- $(1, 1, 1)$
 - Value: 69.141077
 - Error: 8.260978453847002
 - Relative error: 0.106728153953046
- $(1, 2, 2)$
 - Value: 84.146830
 - Error: 0.0000000000000057
 - Relative error: 0.000000000000001
- $(2, 1, 2)$
 - Value: 42.699004
 - Error: 26.667290901163199
 - Relative error: 0.384441619433683

5.2 A_{CU}

- $cu, 2$
 - Value: 38.884836
 - Error: 1.811485408570462
 - Relative error: 0.044512264456278

5.3 A_{HU}

- $hu, 1$
 - Value: 14.091786
 - Error: 0.861079257882436
 - Relative error: 0.057586237455985



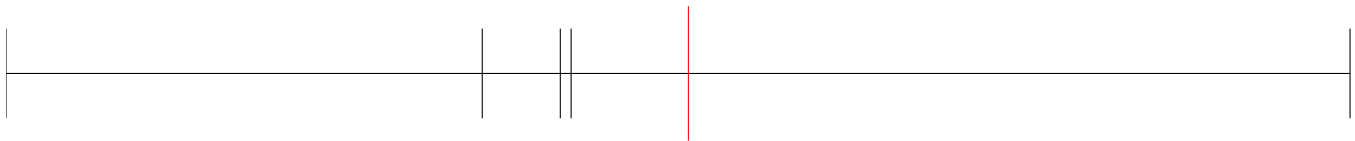
(a) $A^\beta - (1, 1, 1)$



(a) $A^\beta - (1, 2, 2)$



(a) $A^\beta - (2, 1, 2)$



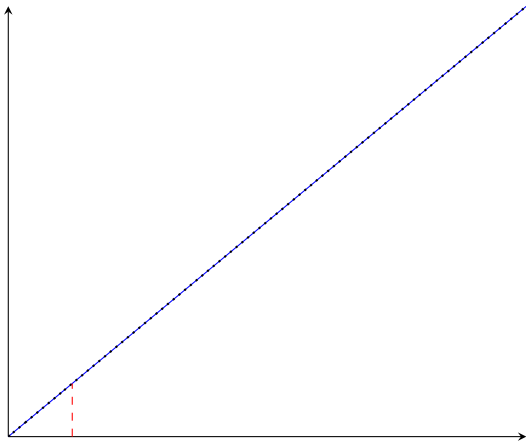
(a) $A^\beta - cu, 2$



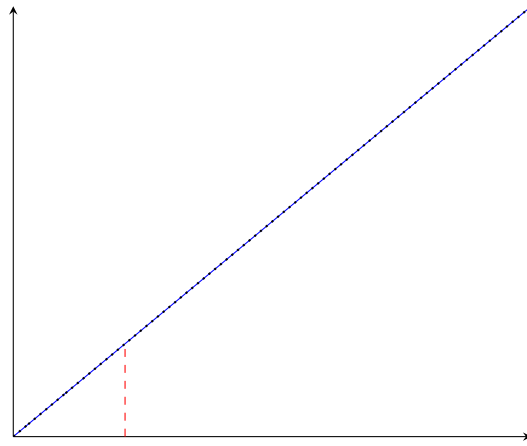
(a) $A^\beta - hu, 1$

6 A^β Breakpoints

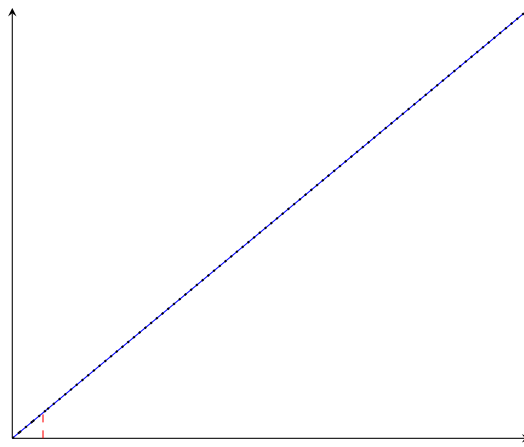
- (1,1,1)
 - q : 69.141077
 - $\hat{q}^\beta$: 69.141077
 - q^β : 69.141077
 - Breakpoints: [0, 5.0, 18.13353447520507, 60.38493803732795, 560.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (1,2,2)
 - q : 84.146830
 - $\hat{q}^\beta$: 84.146830
 - q^β : 84.146830
 - Breakpoints: [0, 11.262728503004022, 39.36524453694061, 390.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- (2,1,2)
 - q : 42.699004
 - $\hat{q}^\beta$: 42.699004
 - q^β : 42.699004
 - Breakpoints: [0, 9.898970403313276, 28.33333333333332, 48.67452649027746, 720.0]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $cu, 2$
 - q : 38.884836
 - $\hat{q}^\beta$: 38.884836
 - q^β : 38.884836
 - Breakpoints: [0, 21.04660840930274, 24.97364426355679, 34.23465128608717, 285.4001551947793]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000
- $hu, 1$
 - q : 14.091786
 - $\hat{q}^\beta$: 14.091786
 - q^β : 14.091786
 - Breakpoints: [0, 4.943755299006499, 12.595096067947068, 13.30871858348944, 237.3002543523117]
 - Error: 0.0000000000000000
 - Relative error: 0.0000000000000000



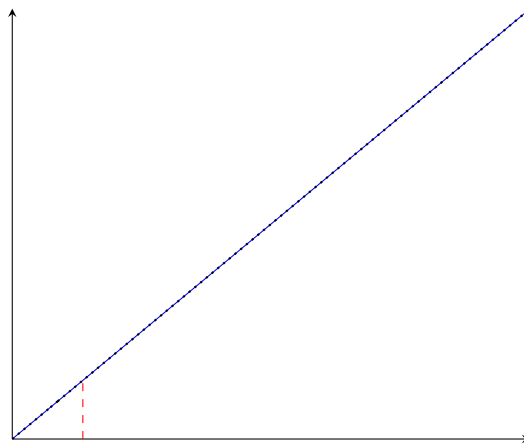
(a) $q^\beta - (1, 1, 1)$



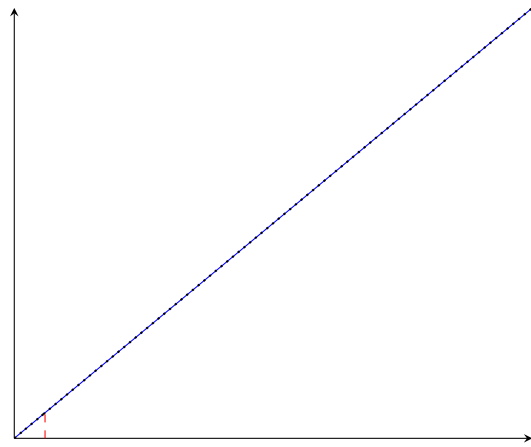
(b) $q^\beta - (1, 2, 2)$



(c) $q^\beta - (2, 1, 2)$



(a) $q^\beta - cu, 2$



7 Other variables

Variable	Value	Variable	Value	Variable	Value
$z_{1,1,2}$	-0.0	$f_{2,2,2}^H$	0.0	$b_{2,2,1}^{H,in}$	0.0
$z_{1,2,1}$	0.0	$f_{1,1,2}^C$	2.2586549421167916	$b_{2,2,2}^{H,in}$	0.0
$z_{2,1,1}$	-0.0	$f_{1,2,1}^C$	13.0	$b_{1,1,2}^{H,out}$	180.24264868023238
$z_{2,2,1}$	0.0	$f_{2,1,1}^C$	0.28732348754047443	$b_{1,2,1}^{H,out}$	4526.785714285715
$z_{2,2,2}$	0.0	$f_{2,2,1}^C$	0.0	$b_{2,1,1}^{H,out}$	11800.0
$z_{cu,1}$	-0.0	$f_{2,2,2}^C$	0.0	$b_{2,2,1}^{H,out}$	0.0
$z_{hu,2}$	-0.0	$RecLMTD_{1,1,2}$	0.1	$b_{2,2,2}^{H,out}$	0.0
$A_{1,1,2}^\beta$	0.0	$RecLMTD_{1,2,1}$	0.008428720083246619	$b_{1,1,2}^{C,in}$	926.0485262678845
$A_{1,2,1}^\beta$	0.0	$RecLMTD_{2,1,1}$	0.029166666666666668	$b_{1,2,1}^{C,in}$	6500.0
$A_{2,1,1}^\beta$	0.0	$RecLMTD_{2,2,1}$	0.012117270959084082	$b_{2,1,1}^{C,in}$	117.80262989159435
$A_{2,2,1}^\beta$	0.0	$RecLMTD_{2,2,2}$	0.1	$b_{2,2,1}^{C,in}$	0.0
$A_{2,2,2}^\beta$	0.0	$RecLMTD_{cu,1}$	0.016124779117675053	$b_{2,2,2}^{C,in}$	0.0
$A_{cu,1}^\beta$	0.0	$RecLMTD_{hu,2}$	0.005549329826813255	$b_{1,1,2}^{C,out}$	926.048526267884
$A_{hu,2}^\beta$	0.0	$q_{1,1,2}$	0.0	$b_{1,2,1}^{C,out}$	6500.0
$A_{1,1,2}$	0.0	$q_{1,2,1}$	0.0	$b_{2,1,1}^{C,out}$	117.80262989159435
$A_{1,2,1}$	0.0	$q_{2,1,1}$	0.0	$b_{2,2,1}^{C,out}$	0.0
$A_{2,1,1}$	0.0	$q_{2,2,1}$	0.0	$b_{2,2,2}^{C,out}$	0.0
$A_{2,2,1}$	0.0	$q_{2,2,2}$	0.0		
$A_{2,2,2}$	0.0	$q_{cu,1}$	0.0		
$A_{cu,1}$	0.0	$q_{hu,2}$	0.0		
$A_{hu,2}$	0.0	$t_{1,1,2}^H$	370.0		
$dt_{1,1,3}$	10.0	$t_{1,2,1}^H$	650.0		
$dt_{1,2,1}$	150.0	$t_{2,1,1}^H$	590.0		
$dt_{2,1,1}$	24.166666666666668	$t_{2,2,1}^H$	590.0		
$dt_{2,2,1}$	90.0	$t_{2,2,2}^H$	370.0		
$dt_{2,2,2}$	31.33528252862242	$t_{1,1,2}^C$	410.0		
$dt_{2,2,3}$	30.992035890340304	$t_{1,2,1}^C$	500.0		
$dt_{cu,1}$	50.0	$t_{2,1,1}^C$	410.0		
$dt_{hu,2}$	180.0	$t_{2,2,1}^C$	350.0		
$f_{1,1,2}^H$	0.48714229373035867	$t_{2,2,2}^C$	350.0		
$f_{1,2,1}^H$	6.964285714285714	$b_{1,1,2}^{H,in}$	180.24264868023238		
$f_{2,1,1}^H$	20.0	$b_{1,2,1}^{H,in}$	4526.785714285715		
$f_{2,2,1}^H$	0.0	$b_{2,1,1}^{H,in}$	11800.0		